

BEYOND BISTABILITY: BIOPHYSICS AND TEMPORAL DYNAMICS OF WORKING MEMORY

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Abstract—Working memory has often been modeled and conceptualized as a kind of binary (bistable) memory switch, where stimuli turn on plateau-like persistent activity in subsets of cells, in line with many *in vivo* electrophysiological reports. A potentially related form of bistability, termed up- and down-states, has been studied with regard to its synaptic and ionic basis *in vivo* and in reduced cortical preparations. Also single cell mechanisms for producing bistability have been proposed and investigated in brain slices and computationally. Recently, however, it has been emphasized that clear plateau-like bistable activity is rather rare during working memory tasks, and that neurons exhibit a multitude of different temporally unfolding activity profiles and temporal structure within their spiking dynamics. Hence, working memory seems to be a highly dynamical neural process with yet unknown mappings from dynamical to computational properties. Empirical findings on ramping activity profiles and temporal structure will be reviewed, as well as neural models that attempt to account for it and its computational significance. Furthermore, recent *in vivo*, neural culture, and *in vitro* preparations will be discussed that offer new possibilities for studying the biophysical mechanisms underlying computational processes during working memory. These preparations have revealed additional evidence for temporal structure and spatio-temporally organized attractor states in cortical networks, as well as for specific computational properties that may characterize synaptic processing during high-activity states as during working memory. Together such findings may lay the foundations for highly dynamical theories of working memory based on biophysical principles. © 2005 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: persistent activity, up-states, neural coding, climbing activity, timing, computational model.

Since the seminal studies by Fuster (1973) and Niki and Kubota (Kubota and Niki, 1971; Niki, 1974), persistently elevated firing rates throughout the delay periods of working memory tasks have been proposed as a neurophysiological correlate for the active maintenance of goal-related information. By now, a wealth of different studies has revealed systematic and stimulus- and/or response-specific changes in single neuron firing rates during the delay

periods of such tasks (Quintana et al., 1988; Funahashi et al., 1989; Goldman-Rakic, 1995; Miller et al., 1996; Rao et al., 1997; Rainer et al., 1998; Fuster et al., 2000; Fuster, 2001; Miller and Cohen, 2001; Diekamp et al., 2002). These studies suggested that neurons carry specific goal-related information across a temporal delay, in the absence of external cues, by enhancing their discharge rate. In fact, spontaneous breakdown of delay-type activity (Funahashi et al., 1989), experimental disruption by either playing distracting sensory stimuli (Fuster, 1973) or by intracortical electrical stimulation (Sobotka et al., 2005), and delay-activity related to the wrong stimulus/response (Quintana et al., 1988), all lead to associated errors in behavioral responses. Fig. 1 reproduces a finding from Funahashi et al. (1989) which summarizes what is thought of to be the essence of the neural basis of working memory: Neurons that switch from a low-activity baseline state into a stimulus-selective high-activity state which they maintain throughout the delay period, after which they return to the low-activity state.

Most commonly it is assumed that this type of neural bistability¹ is brought upon by recurrent neural connections and reverberatory activity, like within a Hebbian cell assembly (Hebb, 1949), where different assemblies might have formed for different stimuli (Amit and Brunel, 1995, 1997; Durstewitz et al., 2000b; Wang, 2001). In neural network models with nonlinear dynamics, the low and high activity states are usually identified as two different attractor states of the network dynamics (Fig. 2), where external stimuli or internal events can switch the networks back and forth between these two stable states (or, more precisely, between the global low and one of the multiple selective high activity states). This sort of bistability is created through an interplay between positive and negative feedback loops (Fig. 2A): As long as the firing rate of a stimulus-selective population of excitatory neurons (pyramidal cells) is low, it will be kept in that low state by feedback inhibition from interneurons which cannot be overcome by occasional small increases in the firing rate of some of the pyramidal cells. However, once activity within the population of pyramidal neurons crosses a certain threshold, e.g. induced by a sensory stimulus, the recurrent excitation within that neural assembly may overcome the feedback inhibition, driving the whole network into the high activity state (Fig. 2B). Once in that state, it will be maintained through the strong feedback excitation alone, i.e. through the positive feedback loop between firing rate and recur-

¹ The term “bistability” will usually be used here in the sense of two stable fixed points of the firing rate.

*Corresponding author. Tel: +44-1752-232711; fax: +44-1752-233349. E-mail address: daniel.durstewitz@plymouth.ac.uk (D. Durstewitz). Abbreviations: ADP, after-depolarizing; CS, conditioned stimulus; PFC, prefrontal cortex; US, unconditioned stimulus.

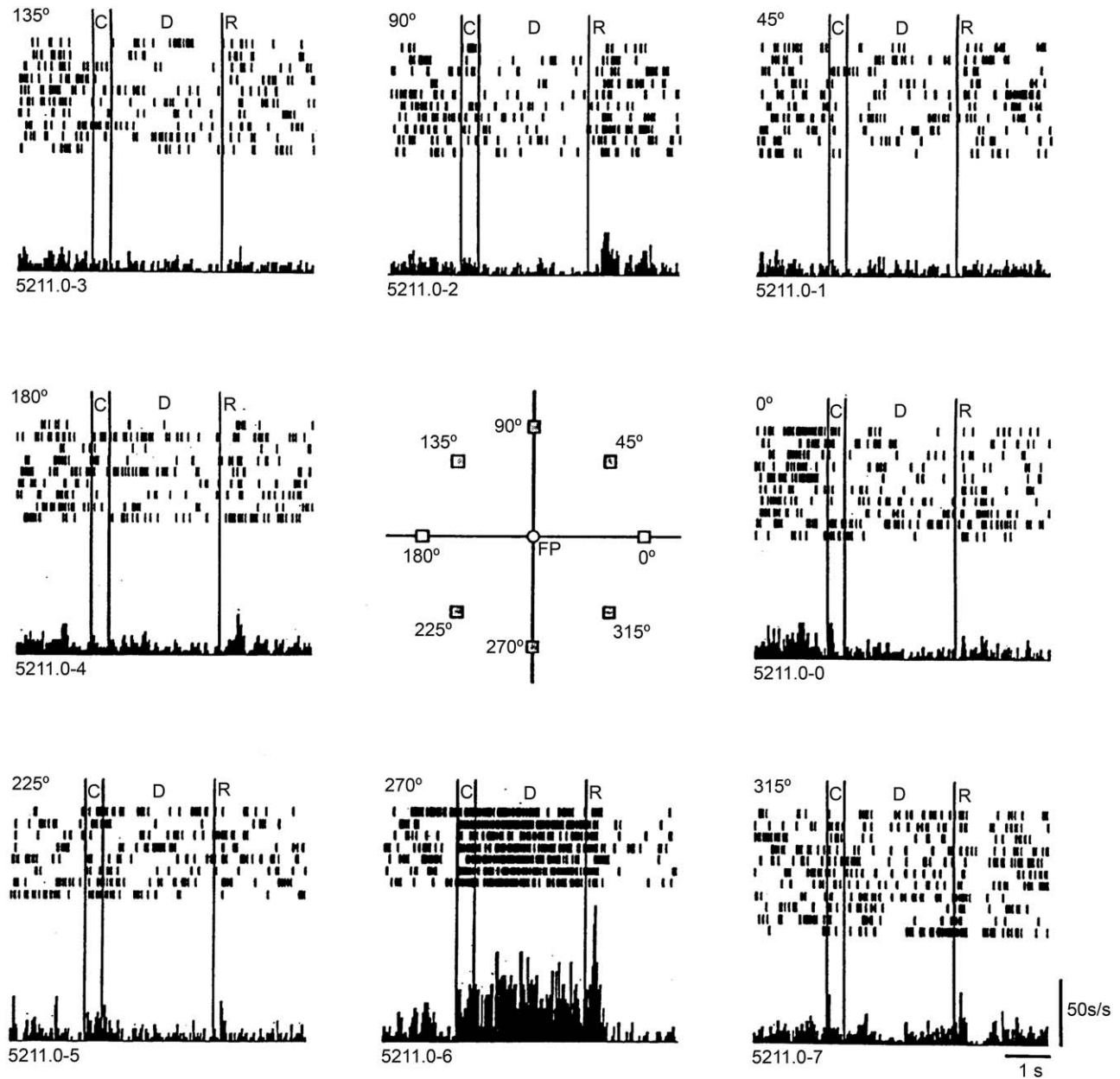


Fig. 1. Stimulus-selective firing rate bistability in a working memory task *in vivo*. A neuron recorded under different stimulus conditions (spatial locations of a cue) in a spatial delayed-matching-to-sample task switches from a low into a high firing rate state after stimulus presentation, and maintains it throughout the delay. This delay-period activity is selective for a cue presented at 270°. Tick marks in the spike raster plots for each of the eight locations give the spike time points, and each row corresponds to a separate trial. The bottom parts of each graph give the spike count histograms. Reprinted from Funahashi et al. (1989), used with kind permission (Copyright 1989 by the American Physiological Society).

rent excitatory synaptic input, even in the absence of external input (Fig. 2C).

Biophysics of bistability

Much modeling and experimental work focused on the question of what precisely the synaptic and cellular mechanisms might be that introduce this sort of bistability, and how this would be modulated through substances like dopamine which play a prominent role in working memory (Sawaguchi and Goldman-Rakic, 1994; Müller et al., 1998; Seamans et al., 1998). Delay-period activity during work-

ing memory is traditionally recorded in behaving animals using extracellular single unit recordings. Such a preparation is not readily amenable to detailed cellular investigations. In order to examine the synaptic and cellular basis of bistability, many recent experimental studies have focused on a potentially related form of membrane bistability, termed up- and down-states, in which the membrane potential of individual neurons can settle at either one of two sustained levels (down: ~ -75 mV, up: ~ -60 mV), associated with very low (essentially zero) and higher firing rates respectively (2–10 Hz) (Fig. 3A). Transitions between

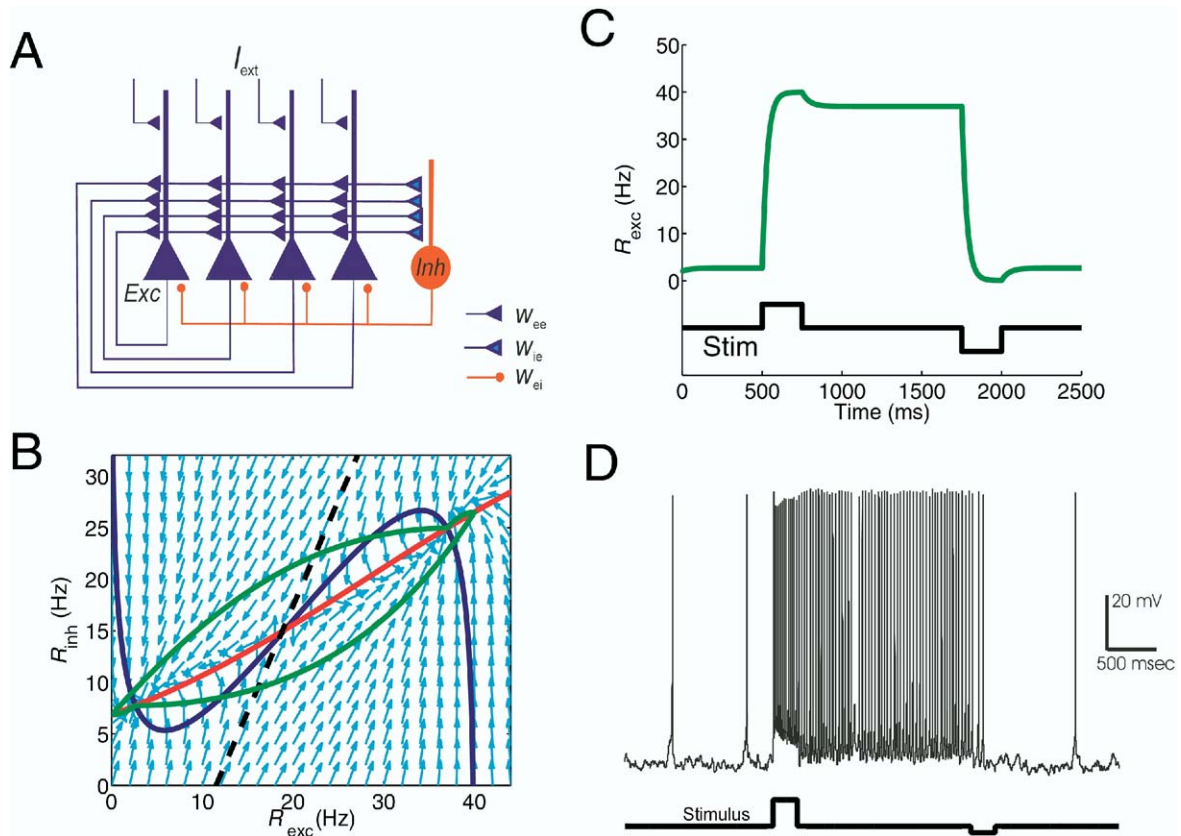


Fig. 2. Firing rate bistability through interaction of positive and negative network feedback loops. (A) Simple model network architecture that may yield bistable behavior: Pyramidal cells (triangles) are recurrently coupled through excitatory synapses (w_{ee}), and also excite (w_{ie}) a population of interneurons (circle) which feeds back inhibition (w_{ei}) to the pyramidal cells. (B) State space of such a model system, depicting the average firing rate of interneurons (R_{inh}) vs. average firing rate of pyramidal cells (R_{exc}). Arrows indicate the flow field, i.e. the direction of change of the average firing rates at various points in the state space. The blue and red lines indicate the points at which the rate of change of firing rate of the pyramidal cells and interneurons, respectively, becomes zero. Intersections of these lines yield three fixed points, at which neither the firing rate of the pyramidal cells nor that of the interneurons changes anymore—the upper and lower ones are stable against small perturbations and hence attractors of the system dynamics, while the middle one is unstable and divides the state space into two basins of attraction (black dashed line). The green curve gives the trajectory of the system corresponding to the time graph in C. (C) An excitatory stimulus switches the system into its high attractor state which is maintained even after offset of the stimulus, and an inhibitory stimulus causes the trajectory to cross the border between the basins of attraction again in the other direction, thus turning persistent activity off. (D) Bistability in a biophysical model network (from a simulation as in Durstewitz and Seamans, 2002).

these states can occur spontaneously, or can be triggered by stimulation, and have been observed in a variety of brain regions *in vivo* during certain stages of the natural sleep–wake cycle, in anesthetized, sometimes also in awake animals, and in reduced cortical slabs, organotypic cultures, or brain slices (Steriade et al., 1993a,b; Plenz and Aertsen, 1996; Plenz and Kitai, 1996; Stern et al., 1998; Lewis and O'Donnell, 2000; Timofeev et al., 2000, 2001; Steriade, 2001; West and Grace, 2002; McCormick et al., 2003; Petersen et al., 2003a,b; Seamans et al., 2003; Shu et al., 2003a; Sachdev et al., 2004).

Empirically observed up-states share important features with recurrent network models of bistable activity (Fig. 2D). High activity states in biophysical network models (i.e. models based on ionic conductances and with spiking dynamics) are—like up-states—accompanied by an increase in both average membrane potential and membrane potential fluctuations, and spiking during these states can be quite irregular (Durstewitz et al., 2000a; Brunel and Wang, 2001). Moreover, up-states both exper-

imentally and in many biophysical models are characterized by a massive increase of especially slow NMDA excitatory and GABA_A inhibitory synaptic currents (Fig. 3B, C; Wang, 1999; Compte et al., 2000; Durstewitz et al., 2000a; Brunel and Wang, 2001; Durstewitz and Seamans, 2002; Seamans et al., 2003; Shu et al., 2003a,b). When prefrontal cortex (PFC) neurons in cultures were voltage-clamped above the chloride reversal potential, >70% of unitary synaptic events were outward during up-states, consistent with GABA_A-mediated inhibitory postsynaptic potentials (IPSPs) (Seamans et al., 2003; Fig. 3C). On the other hand, up-states were driven by a prolonged inward current which could specifically be removed by NMDA antagonists (Fig. 3B). While antagonists of AMPA excitatory currents eliminated the entire up-state, NMDA antagonists eliminated selectively the long-lasting depolarization associated with the up-state while leaving the events which initiate the up-state intact (Seamans et al., 2003).

Modeling has shown that brief AMPA currents alone could be sufficient to maintain persistent activity at physi-

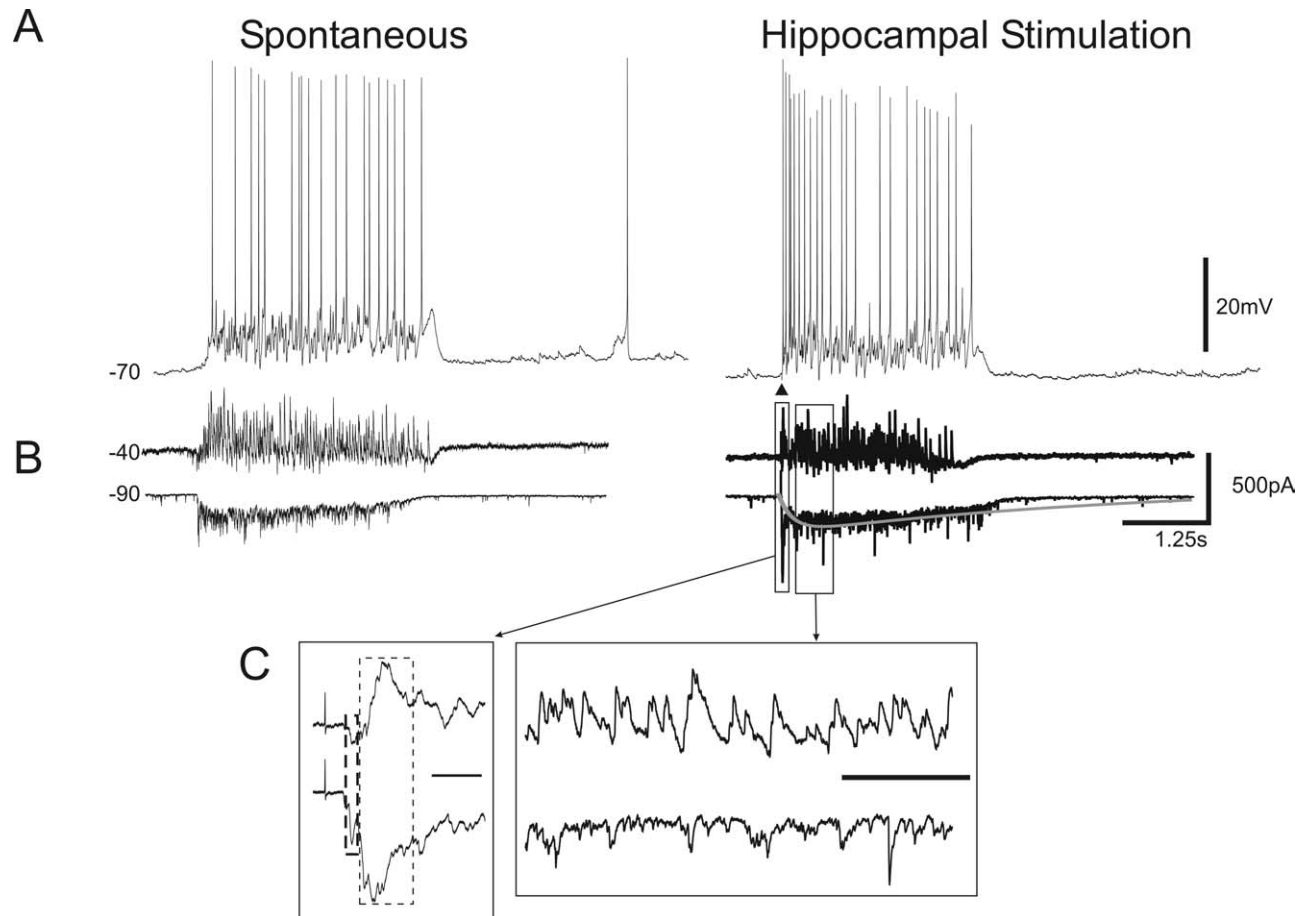


Fig. 3. Up-states in prefrontal cortex (PFC)–midbrain–hippocampal organotypic cultures. A spontaneous up-state (left) and an up-state evoked by hippocampal stimulation (right) recorded in current-clamp (A) and in voltage-clamp (B) from a PFC neuron. In (B) the neuron was voltage clamped to -40 mV (top trace) or -90 mV (bottom trace). The gray line represents a fit to the prolonged inward current observed during the up-state in voltage-clamp. This current was eliminated by an NMDA antagonist. (C) (Left box) An EPSP/IPSP sequence was observed at the start of the up-states. (Right box) Throughout the up-state the cell was bombarded by multiple synaptic events. Note that most events reversed in polarity between -90 mV and -40 mV and were GABA mediated. On average approximately 70% of all such events during up-states were inhibitory. Inset scale bars = 25 ms (left, right 150 ms). Inset scale bars = 25 ms (left), 150 ms (right). See Seamans et al. (2003) for details.

ological rates, provided spiking activity is asynchronous and the network is sufficiently large (Hansel and Mato, 2001; Gutkin et al., 2001; Koulakov, 2001; see also Latham and Nirenberg, 2004). But, unlike AMPA synaptic conductances, NMDA conductances have a long decay time constant and produce a current that can outlast the period of agonist application by as much as 1000-fold at room temperature (Spruston et al., 1995). As a result, these currents are capable of producing sustained depolarizations for seconds. In computational models, this property strongly enhanced the robustness of persistent high activity states at low (far from saturation) firing rates (<30 Hz), even with potentially highly synchronous spiking and small-size cell assemblies (Wang, 1999; Durstewitz et al., 2000a; Koulakov, 2001; for reviews see Durstewitz et al., 2000b, and Wang, 2001). Thus, both biophysical modeling and experimental studies suggest a prominent role for slow excitatory NMDA currents in maintaining up-states, while activity during these states and transitions between states are subject to strong GABA_A-mediated inhibitory control (Amit and Brunel, 1997).

Some authors also have observed *in vitro* a multistability between a global down-state and multiple spatially defined high activity attractor states (Cossart et al., 2003), just as assumed as the basis for stimulus-selective high activity states in the models (Amit and Brunel, 1997; Compte et al., 2000; Durstewitz et al., 2000a,b; Constantinidis and Wang, 2004; Latham and Nirenberg, 2004). However, in many preparations like cortical slabs or during sleep cycles *in vivo*, up-states often take the form of traveling waves, pointing to a more global spatio-temporal dynamics (Timofeev et al., 2000; Petersen et al., 2003a,b; see also Sanchez-Vives and McCormick, 2000).

Neuromodulatory control of bistability

Cortical up-states may also be initiated by stimulation of the hippocampus (Fig. 3A, right) or neuromodulatory mid-brain or brain stem regions associated with arousal (Steriade et al., 1993a; O'Donnell and Grace, 1995; Lewis and O'Donnell, 2000; Seamans et al., 2003). Moreover, the synaptic components that characterize model delay-period

activity and up-states (NMDA and GABA_A) are also specifically targeted in the PFC by dopamine (Seamans et al., 2001a,b). By simulating the dopaminergic modulation of GABA_A and NMDA (and other synaptic and voltage-gated) currents, it was found that dopamine may modulate bistability in an interesting manner: The simultaneous modulation of various synaptic components could lead to a deepening and widening of the basins of attraction of low and high activity states, so that it becomes more difficult to switch between different active memory states (Durstewitz et al., 1999, 2000a; Brunel and Wang, 2001; Durstewitz and Seamans, 2002). This may have interesting functional implications for working memory: Once the network is in a persistent activity state, interfering stimuli or noise has a harder time to push the network out of that state, i.e. active memory states become more robust to distraction (Durstewitz et al., 1999, 2000a; Compte et al., 2000; Brunel and Wang, 2001; Dreher et al., 2002). This effect is similar in nature but not identical to the increase in the signal-to-noise ratio by dopamine postulated previously (Foote et al., 1975; Servan-Schreiber et al., 1990). The associated dopamine-induced changes of low spontaneous and high evoked firing rates have recently also been confirmed experimentally (Lavin et al., 2005), and are in line with the effects of dopamine on delay-period activity *in vivo* (Sawaguchi et al., 1990), and the change in the PFC blood oxygenation level-dependent (BOLD) response following amphetamine administration during working memory tasks in humans (see Seamans and Yang, 2004, or Winterer and Weinberger, 2004 for details).

Relationship of up-states to delay-period activity *in vivo*

Because transitions between down- and up-states also occur in anesthetized animals, during sleep, or in cortical slabs or slices, it is not obvious *per se* how this phenomenon relates to task-related delay-activity in the awake brain during working memory. Strictly, the terms up- and down-state refer to transitions in the average membrane potential, while models of working memory generally speak of bistability in the firing rate. In fact, there are rarely quiescent down-states in the awake cortex (Steriade et al., 2001; Timofeev et al., 2001; but see Petersen et al., 2003a), unlike in anesthetized preparations or acute slices. Both the mean membrane potential during up-states and the average membrane potential associated with quite wakefulness are around -62 mV (Steriade et al., 2001), and average firing rates are both in the range 2–10 Hz, suggesting that mechanisms associated with up-states in anesthetized preparations are more closely related to the low-firing activity state associated with wakefulness in general, rather than with the high-firing activity state that characterizes delay periods.

On the other hand, at least in many biophysical computational models, bistability in membrane potential and bistability in the firing rate are closely related (Lisman et al., 1998; Durstewitz et al., 2000a). Spontaneous firing during the 'down-state' in the biophysical models (Compte et al., 2000; Durstewitz et al., 2000a; Brunel and Wang, 2001) is

usually caused by noisy background (=external, non-recurrent) synaptic inputs which lead to jiggling in the membrane potential with occasional threshold crossings (but see Amit and Brunel, 1997; Brunel, 2000; Wyart et al., 2005), and these background inputs also affect activity during the model 'up-state.' Similar mechanisms might be operational in the awake cortex, with 'background inputs' arising from subcortical, in particular midbrain, structures, but which are lacking in reduced cortical preparations and may be altered during sleep and anesthesia. Moreover, as outlined above, both up-states and delay-period activity in many models critically depend on network interactions, governed by NMDA and GABA synaptic input, and are neither initiated nor maintained by single neuron properties alone (Wang, 1999; Durstewitz et al., 2000a; Rao et al., 2000; Steriade et al., 2001; Seamans et al., 2003; Constantinidis and Wang, 2004). Likewise, delay-period activity *in vivo* and behavior in working memory tasks are strongly disrupted by NMDA antagonists (Dudkin et al., 1997; Shima and Tanji, 1998; Aura and Riekkinen, 1999; but see Lissek and Güntürkün, 2004, for potentially different effects in birds). Hence, studying transitions between down- and up-states may still inform us about biophysical feedback loops driving transitions between attractor states in awake cortex, where low and high activity states are additionally modulated through various 'external' sources of synaptic input.

Up-states could potentially also form a fundamental operational mode of cortical networks, upon which stimulus-specific task-related processing is superimposed. Both during up-states and during task-related processing in the awake brain, neurons receive barrages of synaptic activity (O'Donnell and Grace, 1995; Paré et al., 1998; Stern et al., 1998; Destexhe and Paré, 1999; Lewis and O'Donnell, 2000; Timofeev et al., 2000, 2001; Steriade, 2001; West and Grace, 2002; Seamans et al., 2003; Shu et al., 2003a,b). This profoundly alters their processing properties in a way that may favor various computational operations during delay periods—a point to which we will return later (Bernander et al., 1991; Paré et al., 1998; Destexhe and Paré, 1999; Shu et al., 2003b; Léger et al., 2005).

Cellular bistability

Up-states in organotypic cultures and *in vivo* most often occur synchronously in pairs of nearby cells and quite phase-locked to local field potential oscillations, or take the form of traveling waves reoccurring at regular or irregular intervals (Contreras and Steriade, 1995; Stern et al., 1998; Timofeev et al., 2000; Seamans et al., 2003). Moreover, as described above, they are abolished by blockade of AMPA or NMDA synaptic currents (Seamans et al., 2003). These features suggest that up-states in these preparations are mainly a network property, or at least crucially depend on synaptic input.

While these up-states are predominantly a network phenomenon, established through positive and negative feedback loops that operate through synaptic connections, forms of bistability also exist at the single neuron level. Single neurons are endowed with a variety of internal

positive and negative feedback loops that could produce or augment bistability (Andrade, 1991; Marder et al., 1996; Haj-Dahmane and Andrade, 1997, 1998; Egorov et al., 2002; Loewenstein et al., 2005). For instance, a rise in the intracellular amount of Ca^{2+} activates a depolarizing ionic conductance in entorhinal and prefrontal pyramidal cells (Haj-Dahmane and Andrade, 1997, 1998) which could lead to spiking of the neuron. Spiking in turn fosters the influx of further Ca^{2+} ions into the cell through voltage-gated Ca^{2+} channels present in the soma and dendrites, activating still more of the Ca^{2+} -gated depolarizing channels, thereby further enhancing firing rate and thus closing the positive feedback loop. Such a form of bistability, relying on the interplay of these currents, has been described in entorhinal and PFC pyramidal cells *in vitro*, bathed in the (muscarinic) acetylcholine (ACh)-receptor agonist carbachol which enhances a Ca^{2+} -dependent inward current (Andrade, 1991; Egorov et al., 2002; see Fransén et al., 2002, for a working memory model based on this current). Cellular bistability induced through other ionic mechanisms, like dendritic high-voltage-activated Ca^{2+} or other slowly or non-inactivating inward currents, is also a common feature of many non-cortical preparations and has been proposed to play a role, for instance, in short-term maintenance in the cerebellum which lacks recurrent excitatory connections (Loewenstein et al., 2005).

Cellular bistability has the advantage that it is independent from a specific synaptic matrix—it does not require for stimulus maintenance that strong synaptic links have formed within a cell assembly, and hence it might convey a mechanism for maintaining novel items for which a pre-structured synaptic matrix does not yet exist (Lisman et al., 1998). Whether cellular bistability occurs in the intact neocortex *in vivo*, most likely under certain neuromodulatory or arousal conditions, remains yet to be determined. More generally, *in vivo* any form of cortical bistability is likely to depend on an interplay of synaptic and several intrinsic, voltage- and Ca^{2+} -gated, membrane currents (Tseng and O'Donnell, 2005). The contribution of the latter, however, is still less well explored than that of various synaptic components (see previous section).

Temporal dynamics beyond bistability

Slowly varying activity profiles during delay periods. A variety of data suggests that there is more to delay-period activity than just firing rate bistability. Already Fuster (1973) noted that the activity of many neurons in PFC dynamically changes throughout the course of a delay period, rather than settling into a fixed firing rate. In recent years it has been pointed out that only a minority of neurons may exhibit the clear bistability of the type illustrated in Figs. 1 and 2, basically switching between two plateau-like levels (Rainer and Miller, 2002; Brody et al., 2003). Much more common, the firing rate of many neurons slowly (and consistently) ramps up or down during the delay (Fig. 4A), peaks somewhere in between, or exhibits even more complicated dynamics (Miller et al., 1996; Chafee and Goldman-Rakic, 1998; Quintana and Fuster, 1999; Rainer et al., 1999; Romo et al., 1999; Rainer and

Miller, 2002; Brody et al., 2003). These observations suggest that most neurons do not simply switch from a low into a high firing attractor state after cue presentation, and that during the delay periods there is more going on than just maintenance of information.

One hint to the computational nature of the slowly rising and decaying activity profiles comes from clever experimental designs that allow to separate cue- from response- or target-related neural responses (Quintana and Fuster, 1999; Rainer et al., 1999). Rainer et al. (1999) combined a delayed paired-associate with a delayed-matching-to-sample task where responses of single neurons to the same target but for different cues could be compared. They found that some neurons during the delay are selective for specific cues and not correlated with the target, while others showed responses selective for certain *expected* targets, regardless of the cue. The activity of these latter neurons therefore was anticipatory or prospective in the sense that it was correlated with a stimulus which was yet to be presented. Furthermore, while cue-selective activity tended to decay throughout the delay, target-selective anticipatory activity commonly exhibited a rising profile which peaked around the time of the target presentation. Quintana and Fuster (1999) made similar observations in a quite different design where they manipulated the likelihood with which cues indicated the correct target response to be made after the delay. Hence there seem to be transitions during the delay phase that may indicate a shift from sensory-related processing to preparation of upcoming responses.

These findings are in line with cognitive views on PFC function. The anticipatory nature of the activity is more consistent with prospective coding of mnemonic information, rather than simply serving as a short-term storage function. Indeed, PFC lesions are consistently without effect on traditional short-term memory tasks, such as digit span (Hebb, 1939, 1977; Stuss and Benson, 1986; Petrides, 1989; D'Esposito and Postle, 1999; Manes et al., 2002). Rather the PFC is a region that uses mnemonic information to guide and update forthcoming action which is consistent with the prospective coding at the level of single neurons.

A number of recent observations suggest an additional computational interpretation for climbing activity profiles observed during delays. Animals extract precise temporal intervals between stimuli during classical or operant conditioning and delayed response tasks, and seem to use this information to predict the time of occurrence of forthcoming events relative to preceding predictors (Miller and Barnett, 1993; Staddon and Higa, 1999; Gallistel and Gibbon, 2000; Sargis and White, 2001; Dragoi et al., 2003). A number of ingenious behavioral experiments by Barnett, Cole, and Miller (Cole et al., 1995; Barnett and Miller, 1996; Barnett et al., 1997) yielded results incompatible with purely 'associative' views of conditioning: For example, in one type of experiment, a conditioned stimulus (CS) 1 in a trace conditioning protocol was first paired with a footshock unconditioned stimulus (US) after a delay, and then in separate trials a secondary CS2 was paired with

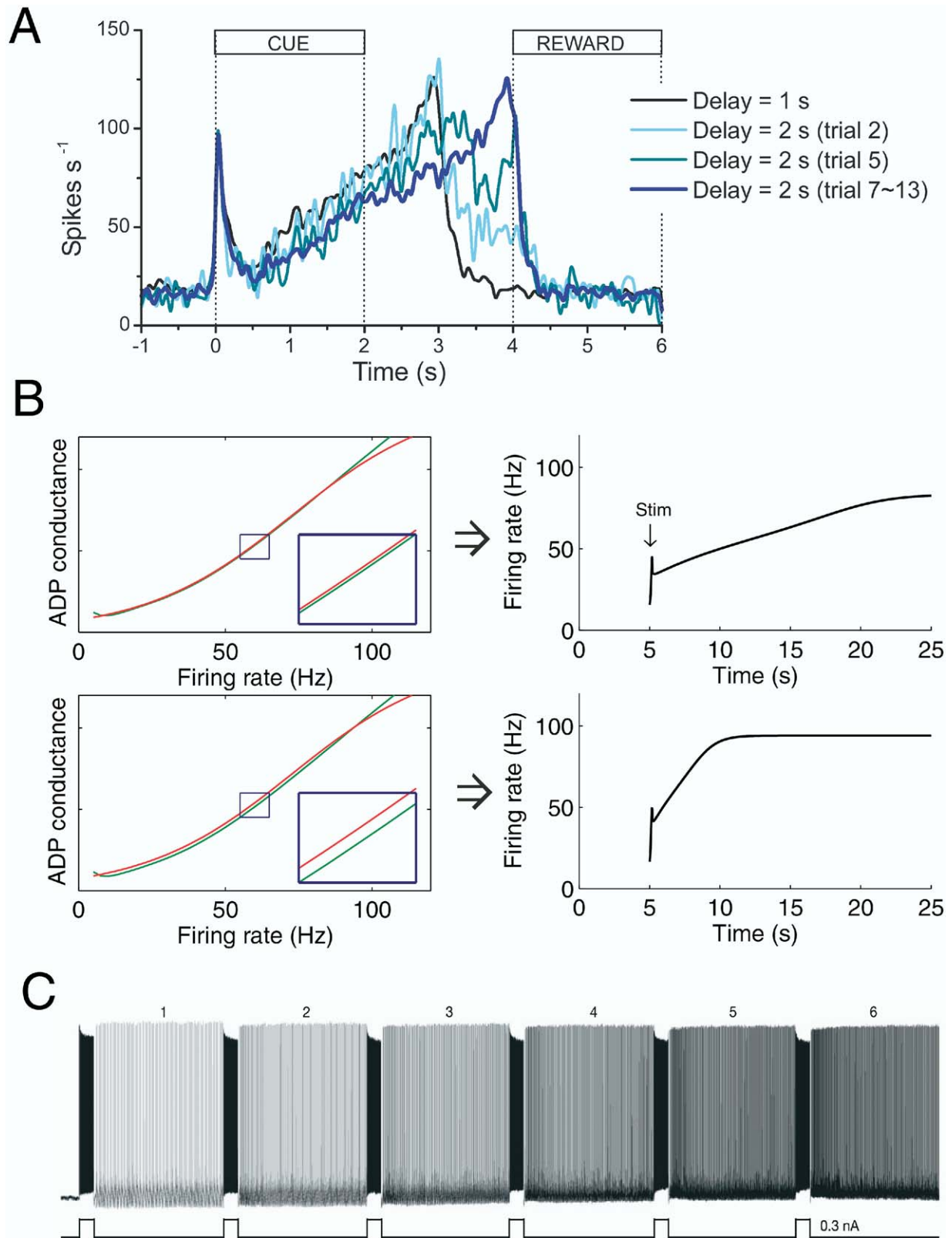


Fig. 4. Mechanisms of adjustable ramping activity. (A) The firing rate of thalamic neurons in a trace conditioning task slowly ramps up during the delay after cue presentation, and peaks at the expected time of occurrence of reward. After changing the length of the temporal interval between cue offset and reward, the slope of climbing activity quickly readjusts within a few trials. Reprinted from Komura et al. (2001) with kind permission (Copyright 2001 by Nature Publishing Group). (B) Dynamical model of climbing activity based on fine tuning of a positive single cell feedback loop between firing rate

CS1 such that the CS2 would have exactly filled the temporal gap between CS1 offset and footshock onset in the former trials. The CS2 finally elicited stronger (suppression of) responses than in subjects where the trace delay between CS1 and US was omitted (i.e. where CS1 offset coincided with US onset), and responses not significantly weaker than to the primary CS1. This was the case although the CS2 itself 1) was never directly paired with the US, and 2) was only backward conditioned to the secondary reinforcer CS1. Apparently the animals had extracted the temporal relations between on- and offset times of the stimuli, and had done the arithmetic to infer the time of occurrence of the US relative to the CS2. Some psychologists now believe that extracting temporal relations and precise interval times between events is one of the most fundamental learning processes (Miller and Barnet, 1993; Staddon and Higa, 1999; Gallistel and Gibbon, 2000). From an ecological point of view, this makes sense, as anticipatory behavior is most useful if somehow time-locked to the anticipated events.

Such behavioral findings put working memory tasks into a broader context as they suggest that animals may be extracting temporal intervals incidentally, even if not required for the task at hand. If animals do not only anticipate targets that are due to appear after a delay (Rainer et al., 1999) but in addition anticipate their time of occurrence as a fundamental part of working memory processing, the question becomes: How is interval time encoded within the activity of single neurons? A number of experimental observations suggest that slowly ramping activity profiles during the delay period may in fact carry interval time information. Komura et al. (2001) in a landmark study using a trace conditioning task found that thalamic neurons exhibited a remarkably linearly rising activity after cue onset and throughout the delay between cue offset and reward (Fig. 4A). Activity started at about ~20 Hz and peaked at ~120 Hz at the time of reward presentation. Most interestingly, the slope of this climbing activity adjusted to the length of the delay within just a few trials! If the delay was increased from 1 s to 2 s, the rate of change of climbing activity slowed down until it peaked again with the same maximum firing rate at the new time of occurrence (Fig. 4A). Vice versa, if the delay interval was shortened, climbing activity sped up. These findings show that the slope of climbing activity is related to the temporal interval employed, and hence may carry information about interval time.

Climbing activity has been observed by a number of groups in prefrontal and other neocortical areas during working memory and association tasks (Sakai and Mi-

yashita, 1991; Miller et al., 1996; Chafee and Goldman-Rakic, 1998; Quintana and Fuster, 1999; Rainer et al., 1999; Romo et al., 1999; Brody et al., 2003), and recently it was found that its slope just like in thalamus scales with delay period length in working memory tasks (Brody et al., 2003; Reutimann et al., 2004). Early experimental evidence that neurons in PFC may encode temporal intervals comes from Niki and Watanabe (1979). Taken together, these studies revealed a number of properties of slowly ramping activity that make it a good candidate for processing interval time during working memory (reviewed in Durstewitz, 2004): 1) Ramping activity may span a wide range of different intervals ranging from hundreds of milliseconds (Rainer et al., 1999) to tens of seconds (Quintana and Fuster, 1999). 2) Ramping activity is often surprisingly linear as one may expect from a perfect temporal integrator (Komura et al., 2001). 3) Ramping activity adjusts to the length of the temporal interval employed in the task (Komura et al., 2001; Brody et al., 2003). A recent study by Leon and Shadlen (2003) adds to these findings more direct evidence for a correlation between psychophysical measures of interval time comparisons and the slopes of rising and decaying activity in parietal cortex.

Computational models of slowly ramping activity. Whatever the functional implications, findings on slowly climbing and decaying activity can be reproduced with models that exhibit simple firing rate bistability only if certain additional assumptions are made. Mongillo et al. (2003) provided one such model where some populations of neurons (cell assemblies) code for different cue stimuli and others for associated targets (see Kitano et al., 2003, for related ideas based on cellular bistability). The basic idea is that there are stronger excitatory synaptic connections from the cue to the associated target assemblies than vice versa, but that in general different cell assemblies inhibit each other indirectly through interneurons. Cues will push cue-selective assemblies into their high attractor states (just as in Figs. 1 and 2), but due to noise in the system and the excitatory connections from cue- to target-selective neurons, a transition may occur where the associated target-selective population pops into its high state while the cue-related population may go down. The likelihood of this transition and hence the average waiting time until its occurrence will depend on the strength of various synaptic connections, in particular the ones from cue- to target-selective neurons. Because transitions due to the noise occur at various times during the delay, averaging over many trials yields climbing activity by smoothing the many temporally jittered binary jumps of bistable activity.

and Ca^{2+} -dependent after-depolarizing (ADP) currents. The red lines in the left hand side graphs give the magnitude of ADP conductance produced at different firing rates, while the green lines give the magnitude that would be required to maintain the neuron at a given rate. Since the strength of depolarizing conductance actually produced is greater than the one required for a range of rates, the firing rate increases over time as shown in the time graphs on the right hand side. However, the rate of change of climbing activity depends on the magnitude of the discrepancy between ADP conductance produced and required—if it is very small (top graphs), activity will rise much slower than commanded by the intrinsic time constants of the system. The tuning of such a feedback loop may establish itself in a self-organizing manner (Durstewitz, 2003). Modified from Durstewitz (2003) with permission (Copyright 2003 by the Society for Neuroscience). (C) Multistability in an entorhinal pyramidal cell in the absence of synaptic input. The cell is switched into different self-sustaining firing rates by depolarizing current steps. Reprinted from Egorov et al. (2002) with kind permission (Copyright 2002 by Nature Publishing Group).

As the average waiting time depends on the strength of the synaptic connections from cue- to target neurons, adjusting these strengths will adjust the slope of climbing activity. Hence, this model suggests that “climbing activity” is simply due to the process of averaging single neuron and trial data in *in vivo* experiments. This may not be the case however, as some authors have observed or provided statistics for climbing activity within single neurons in single trials (Romo et al., 1999; Leon and Shadlen, 2003).

Alternative models for slope-adjustable climbing activity depart from the simple bistability concept (Durstewitz, 2003; Reutimann et al., 2004). One such idea proposes that slowly climbing or decaying activity with variable slopes is not due to switching between two point attractors, but emerges in the vicinity of a so-called line attractor configuration (Fig. 4B; Durstewitz, 2003, 2004). A line attractor is composed of an (ideally) continuous set of stable fixed points, so that the neural system can maintain arbitrary firing rates in a graded fashion. On the biophysical level, line attractors could be implemented through different network (Seung et al., 2000; Koulakov et al., 2002) or single neuron (Durstewitz, 2003; Loewenstein and Sompolinsky, 2003) mechanisms (line attractors could also support decision processes in parametric working memory: Machens et al., 2005). Climbing activity emerges at a bifurcation where the whole line attractor just loses stability (Fig. 4B). Because of that, the system has to travel through a continuous series of states whose former status as true attractors is still ‘felt’ by the system dynamics: It strongly slows down while traveling along the former line attractor, in fact with a speed that could, at least theoretically, be made arbitrarily slow (Durstewitz, 2003). The distance to the line attractor configuration and hence the slope of climbing activity could be regulated through the adjustment of synaptic input.

One experimental observation relevant to this model of climbing activity is that of cellular multistability in entorhinal pyramidal cells *in vitro* (Egorov et al., 2002). Single pyramidal cells in the absence of synaptic input could be switched into multiple self-sustaining firing rates by series of depolarizing or hyperpolarizing current steps (Fig. 4C). This cellular multistability depended on Ca^{2+} influx and Ca^{2+} -activated inward currents, and indicates that the feedback loop between cell firing, Ca^{2+} influx, and Ca^{2+} -activated currents may be finely tuned toward a line attractor configuration. In fact it is the tuning of this feedback loop that underlies the emergence of climbing activity in the model described above (Durstewitz, 2003).

Another model of slowly and adjustable ramping activity exploits a cellular property of interneurons, namely slow spiking adaptation with a time course dependent on the strength of the initiating input (Reutimann et al., 2004). The different slopes of interneuron adaptation are transformed into different slopes of rising activity through interaction with pyramidal cells. Thus, in summary, a number of cellular and network mechanisms have been proposed that could generate ramping profiles as empirically observed. These models make specific predictions that may be used

to differentiate between them experimentally (see Durstewitz, 2003, 2004; Reutimann et al., 2004).

Temporal structure in delay period activity. Another set of findings for which the simple view of working memory as bistable activity does not account for concerns temporal structure in neural activity. Human EEG studies have observed an increase in power in the gamma and theta band during working memory tasks (Raghavachari et al., 2001; Lutzenberger et al., 2002), where the power in the gamma band seems to correlate with working memory load (Howard et al., 2003). Such increases in theta and gamma power were also noticed in local field potential recordings from the monkey parietal and visual cortex specifically during the delay period, and selectively for the preferred stimulus of a recorded neuron (Pesaran et al., 2002; Lee et al., 2005). Some authors observed enhanced gamma frequency oscillations even at the single neuron level (Pesaran et al., 2002; but see Compte et al., 2003 who failed to do so). These findings show that working memory tasks evoke cooperative population phenomena like oscillations specifically during the delay period, and tuned to specific stimuli.

Lee et al. (2005) in addition observed that the activity of single neurons during a delayed response task is often phase-locked to the ongoing theta oscillation, and that this phenomenon may occur in the absence of general increases in firing rate. In other words, these neurons settle into a temporally coordinated pattern in a stimulus-specific manner during delay processing, but show no signs of firing rate bistability. Precisely re-occurring spiking (activity) patterns within single or among multiple neurons that were not directly associated with oscillatory processes have also been observed in frontal and sensory cortices (Riehle et al., 1997; Tsodyks et al., 1999; Fellous et al., 2004). For instance, Riehle et al. (1997) described that two or more neurons precisely aligned their spiking times immediately preceding or following significant task events in frontal cortex during a delay task (see also computational studies by Diesmann et al., 1999; Kitano et al., 2003).

Significance of temporal structure for coding and computation. Neural spiking phase-locked to ongoing oscillations, and spike time correlations between neurons, as have also been measured during delay periods (Vaadia et al., 1995; Funahashi and Inoue, 2000; Constantinidis and Goldman-Rakic, 2002), have been proposed as possible coding mechanisms (von der Malsburg, 1981; Hopfield, 1995; Lisman and Idiart, 1995). In general this is a wide field far beyond the scope of this review, so here we will introduce just a few ideas potentially relevant to working memory processing.

A model by Lisman and Idiart (1995) suggests that multiple item working memory might be neurally implemented by having spikes encoding different items associated with different phases of the theta rhythm (and different gamma subcycles). Temporal sequences might be coded in a similar manner as a phase code (Lisman, 1999; Jensen and Lisman, 2005), and the phase precession effect observed in hippocampal place cells while rats tra-

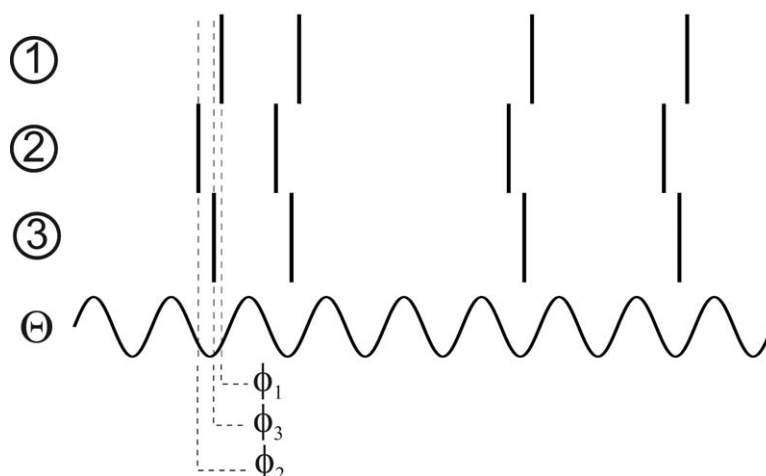


Fig. 5. Phase coding. Spiking pattern of three neurons (circles labeled 1–3) that stereotypically repeats itself several times at irregular intervals, but always phase-locked to an underlying theta population oscillation (as measured, e.g. by the local field potential). Information may be encoded in this spiking pattern as a phase-vector (ϕ_1, ϕ_2, ϕ_3), where the ϕ_i denote the relative position (phase) within the theta cycle. This pattern could be read out by a coincidence detector if the axonal delays $\Delta ax_2 > \Delta ax_3 > \Delta ax_1$ are such that they compensate for the time differences within the theta cycle. This property may establish itself in a self-organizing selection process (Gerstner et al., 1996).

verse different places provides some evidence for such a coding scheme (Jensen and Lisman, 1996).

Hopfield and Brody (Hopfield, 1995; Brody and Hopfield, 2003) suggested an ingenious model of scale-invariant object coding that exploits phase-locked activity. In one version of the model (Hopfield, 1995) the relative strength of activation of different receptors or afferent neurons is translated into a phase code (Fig. 5), where different spikes making up the representation of an object are associated with different phases of a theta cycle. Rescaling the object may change the position of the whole spiking pattern within the theta cycle but may keep the temporal relations between spike times unchanged. Such a spike time representation could be read out by a coincidence detector (Softky and Koch, 1993; König et al., 1996), assuming that neurons firing at different phases have different axonal distances to the coincidence detector neuron such that all spikes arrive simultaneously. Alternatively, different representational neurons may receive additional current inputs that compensate for mismatches in their activation induced by the stimulus, and thereby yield phase-locking with a lag of zero, i.e. precisely synchronous spiking (Brody and Hopfield, 2003). (Both these mechanisms could establish themselves in a self-organizing manner: Gerstner et al., 1996; Brody and Hopfield, 2003.) Precisely synchronized spiking is also the major ingredient of von der Malsburg's (1981) idea where spikes belonging to the same object representation are precisely aligned in time, while spiking times associated with different objects are asynchronous (Singer and Gray, 1995).

Coincidence detection and correlated activity was also suggested as a mechanism that could give rise to the high spiking irregularity of neocortical neurons (Softky and Koch, 1993; Stevens and Zador, 1998; Salinas and Sejnowski, 2000; but irregularity could also result from asynchronous chaotic activity; see van Vreeswijk and Sompolinsky, 1998, and Durstewitz and

Gabriel, submitted). Indeed, if neural activity in PFC is resolved at the single spike level, most neurons exhibit either highly irregular spiking or burst-like activity (Compte et al., 2003), where the latter could go in hand with theta-phase-locked activity (Lee et al., 2005).

Some of the existing bistable working memory models may reproduce aspects like highly irregular spiking (Compte et al., 2000; Durstewitz et al., 2000a; Brunel and Wang, 2001), oscillatory (Compte et al., 2000), or correlated activity (Gutkin et al., 2001), but they usually do not address the computational significance of these phenomena, as they were mainly intended to account for the biophysics of bi- (multi-) stability (but see Lisman and Idiart, 1995; Jensen and Lisman, 1998). The findings on temporally structured and coordinated activity within delay periods, possibly even in the absence of firing bistability, make clear that there is still much more to working memory processing. None of these findings, or of the coding schemes outlined above, however, is *specific* for working memory, as is probably also true for bistability and climbing activity which have been observed in various cortical and subcortical areas. Rather these phenomena may represent neurocomputational universals. Therefore, one task of future modeling efforts may be to further understand the specific role of these phenomena in working memory.

Implications of up-states for neural computations beyond bistability. As we have argued, the neural processes that come into action during delay periods of working memory tasks often might not be adequately or fully described through bistability concepts. However, up-states share important features with delay period activity that go beyond bistability. Most importantly, both states constitute 'high-conductance states' (Destexhe et al., 2003) with a unique contribution and barrage of various voltage-gated and synaptic conductances. Hence, preparations like cultures or slices in which biophysical properties can be more

easily accessed and manipulated than *in vivo*, may be useful to reveal some important computational properties of such states.

One of the most important consequences of the barrage of inhibitory and NMDA-mediated inputs is the strong reduction in the effective membrane time constant of the neurons (Bernander et al., 1991; Paré et al., 1998; Destexhe and Paré, 1999; Léger et al., 2005), i.e. the time constant with which synaptic inputs are integrated, leading to a significant reduction in both the amplitude and duration of excitatory postsynaptic potentials (EPSPs) (Léger et al., 2005). Because of this, the temporal precision with which a neuron responds and its sensitivity to correlated inputs are enhanced, as temporally more precisely coordinated inputs are necessary to make the neuron spike (Softky, 1994; Fellous et al., 2003; Rudolph and Destexhe, 2003). The high spiking irregularity and increased membrane potential variance during high-conductance states and delay-activity may be a signature of such a coincidence detection mode (Softky and Koch, 1993; Stevens and Zador, 1998; Salinas and Sejnowski, 2000; Benucci et al., 2004; but could also reflect other deterministic processes associated with high-conductance states: Durstewitz and Gabriel, submitted). Another consequence of the increased membrane potential variance can be a higher sensitivity to small signals (Hô and Destexhe, 2000; Destexhe et al., 2001; Fellous et al., 2003; Shu et al., 2003b). At the same time spiking responses to strong inputs can be reduced, such that the combined effect of the increases in membrane conductance and conductance noise results in a lowering of the gain of the neurons (i.e. a reduction in the slope of the neural input/output function; Chance et al., 2002; Fellous et al., 2003; Shu et al., 2003b). The reduced gain may enhance the operational range of the neurons with regards to different input firing rates. Hence, high-conductance states have dramatic implications for neural information processing like increased temporal precision, increased sensitivity to small and high-frequency inputs, and a larger operational range (see Destexhe et al., 2003 for further implications). These special properties may partly account for precise temporal structure and a variety of smoothly changing activity profiles observed in working memory tasks.

In fact, precise repetition of temporal sequences has also been observed *in vitro* and in organotypic cultures (Fellous et al., 2004). Cossart et al. (2003) and Ikegaya et al. (2004) showed a form of up-state in an acute brain slice, during which groups of neurons exhibited repetitive (non-random) and sometimes hierarchically structured activation sequences. These findings fit with the 'synfire chain' idea which refers to the propagation of precisely aligned spiking packages through chains of connected neurons (Abeles, 1991; Abeles et al., 1993; Diesmann et al., 1999). Unlike up-states *in vivo*, relatively few neurons at a time participated in these up-states. However, precise firing patterns were most likely during up-state periods, and were abolished by NMDA antagonists. In a landmark study, Beggs and Plenz (2003, 2004) observed that events of cortical activity in neocortical slices and organotypic

cultures obey a power law with slope $-3/2$. Such power law behavior indicates scale-invariance with events happening at all length-scales (in this case, the ratio between the probabilities of observing an event (avalanche) of size $k \times s_0$ and one of size s_0 was independent of size (scale) s_0). This power law with an exponent of $-3/2$ suggests a well-balanced network dynamics or architecture that would support stable propagation of synfire packages without fading or growing of activity. Hence, these studies revealed properties of neocortical networks that could promote precise activity patterns as observed in delay tasks (Riehle et al., 1997).

CONCLUSIONS

In the past 5–10 years, a number of computational studies and reduced experimental preparations like cultures or brain slices have revealed biophysical mechanisms that could establish bistability on the single cell or network level. The hysteresis property inherent in bistability is the classical mechanism to store short-term information. However the role of bistability in working memory is not really that clear. Most neurons recorded in working memory tasks show more intriguing temporally unfolding profiles than simply switching between two stable firing rates. The computational significance of these dynamical observations is not yet clear, but they may be related to the fact that working memory is much more an active process rather than simple maintenance of short-term information. Animals extract the temporal structure of their environment, predict forthcoming events and their time of occurrence, and schedule actions ahead. These processes may be reflected in the slowly ramping activity profiles and transitions observed in neural activity.

Moreover, neural activity during delay periods often seems to be temporally structured at the level of single spikes, just like it is in many other perceptual, motor, or memory processes. Despite or because of this complexity, reduced experimental preparations did a good job in identifying biophysical factors and feedback loops that might govern these types of neural dynamics. In particular, high activity states during the delay period correspond to high-conductance states with a number of unique computational properties as revealed through *in vitro*, *in vivo*, and modeling studies. They favor precise temporal structure, finely graded firing rate responses, and high sensitivity to weak inputs. Moreover, studies *in vitro* and in cultures suggest that neocortical networks are tuned to support stable propagation of precise spiking patterns. These features may help to explain the intriguing temporal dynamics observed in working memory tasks, and may help to develop a theory of working memory anchored in cellular biophysics.

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